usepackage[a4paper,margin=0.65in]geometry

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Contents

556. On Steiner's Surface (continued)

[Continued from earlier pages of the same paper.]

[p. 6] where the first factor gives the node. Equating to zero the second factor, we have

$$\left\{qZ + \left(2q - 1 - \tfrac{2}{q}\right) X\right\}^2 = X^2 \left\{\left(2q - 1 - \tfrac{2}{q}\right)^2 - 4q^2 + 4q - 1\right\}$$

$$= X^2 \cdot \tfrac{4}{q^2}(1-q)(1+2q);$$

or, finally,

$$qZ = \left\{-2q + 1 + \tfrac{2}{q} \pm \tfrac{2}{q}\sqrt{(1-q)(1+2q)}\right\} X,$$

giving two real values for all values of q from $q = 1$ to $q = -\frac{1}{2}$. (See the Table afterwards referred to.)

We may recapitulate as follows:

$q > 1$, or $\lambda > \frac{1}{2}h$; the curve is imaginary, but with three real acnodes, answering to the acnodal parts of the nodal lines:

$q = 1$, or $\lambda = \frac{1}{2}h$; the summit appears as a fourth acnode:

$q < 1 > \frac{1}{4}$, or $\lambda < \frac{1}{2}h > \frac{1}{3}h$; the curve consists of three acnodes and a trigonoid lying within the triangle and having the sides of the triangle for bitangents of imaginary contact:

$q = \frac{1}{4}$, or $\lambda = \frac{1}{3}h$; the curve consists of three acnodes and a trigonoid having the sides of the triangle for osculating tangents:

$q < \frac{1}{4} > 0$, or $\lambda < \frac{1}{3}h > \frac{1}{4}h$; the curve consists of three conjugate points and an indented trigonoid having the sides of the triangle for bitangents of real contact:

$q = 0$, or $\lambda = \frac{1}{4}h$; curve has the summits of the triangle for cusps:

$q < 0 > -\frac{1}{4}$, or $\lambda < \frac{1}{4}h > \frac{1}{6}h$; curve has three crunodes, or say it is a cis-centric trifolium:

$q = -\frac{1}{4}$, or $\lambda = \frac{1}{6}h$; curve has a triple point, or say it is a centric trifolium:

$q < -\frac{1}{4} > -\frac{1}{2}$, or $\lambda < \frac{1}{6}h > 0$; curve has three crunodes, or say it is a trans-centric trifolium:

$q = -\frac{1}{2}$, or $\lambda = 0$; curve is a two-fold circle:

$q < -\frac{1}{2}$, or $\lambda < 0$; the curve becomes again imaginary, consisting of three acnodes answering to the acnodal parts of the nodal lines.

For the better delineation of the series of curves, I calculated the following Table, wherein the first column gives a series of values of $\lambda : h$; the second the corresponding values of $q, = \dfrac{2\lambda - h}{2\lambda - 2h}$; the third the positions of the point of contact, say with the side $Z = 0$, the value of $X : Y$ being calculated from the foregoing formula

$$\frac{X}{Y} = \frac{1}{2q} + 1 \pm \sqrt{\frac{1}{4q^2} - \frac{1}{q}};$$

[p. 7] and the fourth the apsidal distances, say for the radius vector $X = Y$, the value of $Z : X$ being calculated from the foregoing formula

$$\frac{Z}{X} + \frac{X}{Z} = -2 + \frac{1}{q} + \frac{2}{q^2} \pm \frac{2}{q}\sqrt{\left(\frac{1}{q} - 1\right)\left(2 + \frac{1}{q}\right)}.$$

The Table is:

$\lambda : h$	q	Contact, $Z = 0$, $X : Y =$	Apses, $X = Y$; $X : Z =$
$*\ \cdot 75$	$1 \cdot 00$	imposs.	$1\cdot$
$\cdot 70$	$\cdot 6666$	imposs.	$\cdot 032$ or $7 \cdot 968$
$*\ \cdot 666$	$\cdot 5$	$1\cdot$	$0\cdot$ $16\cdot$
$\cdot 65$	$\cdot 4285$	$\cdot 320$ or $3 \cdot 124$	$\cdot 006$ $22 \cdot 44$
$\cdot 6$	$\cdot 25$	$\cdot 0721$ $13 \cdot 9279$	$\cdot 059$ $67 \cdot 941$
$\cdot 55$	$\cdot 1111$	$\cdot 011$ $78 \cdot 988$	$\cdot 15$ $337 \cdot 85$
$*\ \cdot 5$	$0\cdot$	0 or ∞	$\cdot 25$ ∞
$\cdot 45$	$- \cdot 0909$	$\cdot 005$ $118 \cdot 99$	$\cdot 39$ $457 \cdot 61$
$\cdot 4$	$- \cdot 1666$	$\cdot 029$ $33 \cdot 971$	$\cdot 496$ $127 \cdot 504$
$\cdot 35$	$- \cdot 2308$	$\cdot 060$ $16 \cdot 72$	$\cdot 648$ $61 \cdot 79$
$*\ \cdot 3$	$- \cdot 2857$	$\cdot 099$ $10 \cdot 151$	$\cdot 812$ $37 \cdot 187$
$*\ \cdot 25$	$- \cdot 3333$	$\cdot 141$ $6 \cdot 854$	$1\cdot$ 25
$\cdot 2$	$- \cdot 375$	$\cdot 207$ $4 \cdot 904$	$1 \cdot 218$ $17 \cdot 89$
$\cdot 15$	$- \cdot 4118$	$\cdot 276$ $3 \cdot 622$	$1 \cdot 48$ $13 \cdot 25$
$\cdot 10$	$- \cdot 4444$	$\cdot 372$ $2 \cdot 690$	$1 \cdot 813$ $9 \cdot 937$
$\cdot 05$	$- \cdot 4737$	$\cdot 515$ or $1 \cdot 941$	$3 \cdot 15$ or $6 \cdot 46$
$*\ 0$	$- \cdot 5$	$1\cdot$	$4\cdot$

where the asterisks show the critical values of $\lambda : h$.

It is worth while to transform the equation to new coordinates X', Y', Z' such that $X' = 0$, $Y' = 0$, $Z' = 0$ represent the sides of the triangle formed by the three nodes. Writing for shortness $X + Y + Z = P$, $YZ + ZX + XY = Q$, $XYZ = R$, the equation is

$$(qP - Q)^2 = 4(2q + 1)PR.$$

The expressions of X, Y, Z in terms of the new coordinates are of the form $X' + \theta P'$, $Y' + \theta P'$, $Z' + \theta P'$, where $P' = X' + Y' + Z'$; writing also $Q' = Y'Z' + Z'X' + X'Y'$, $R' = X'Y'Z'$, then the values of P, Q, R are

$$(1+3\theta)P', \quad Q' + (2\theta + 3\theta^2)P'^2, \quad R' + \theta P'Q' + (\theta^2+\theta^3)P'^3,$$

[p. 8] and the transformed equation is

$$\left[\{q(1+3\theta)^2 - 2\theta - 3\theta^2\}P'^2 - Q'\right]^2 = 4(2q+1)(1+3\theta)\{(\theta+\theta^2)P'^3 + \theta P'Q' + R'\},$$

which is satisfied by $Q' = 0$, $R' = 0$, if only

$$\{q(1+3\theta)^2 - 2\theta - 3\theta^2\}^2 = 4(2q+1)(1+3\theta)(\theta^2+\theta^3),$$

or, if for a moment $q(1+3\theta) = \Omega$, the equation is

$$(\Omega^2 - 2\theta - 3\theta^2)^2 = 4(\theta^2+\theta^3)(2\Omega + 1 + 3\theta),$$

that is,

$$\Omega^4 + \Omega^2(-6\theta^2 - 4\theta) + \Omega(-8\theta^3 - 8\theta^2) - 3\theta^4 - 4\theta^3 = 0,$$

that is,

$$(\Omega+\theta)^2(\Omega^2 - 2\theta\Omega - 3\theta^2 - 4\theta) = 0.$$

If the new axes pass through the nodes, then $\Omega + \theta = 0$; that is, $q(1 + 3\theta) + \theta = 0$, which equation gives the value of θ for which the new axes have the position in question; substituting in the first instance for q the value $\dfrac{-\theta}{3\theta+1}$, the equation becomes

$$\theta^2\{2\theta(1+\theta)P'^2 + Q'\}^2 = 4(1+\theta)P'\{\theta^2(1+\theta)P'^3 + \theta P'Q' + R'\},$$

that is,

$$Q'^2 = 4(1+\theta)P'R';$$

or, finally, substituting for θ its value in terms of q, the required equation is

$$Q'^2 = 4 \cdot \frac{2q+1}{3q+1} P'R',$$

that is,

$$(Y'Z' + Z'X' + X'Y')^2 = 4 \cdot \frac{2q+1}{3q+1} X'Y'Z'(X'+Y'+Z').$$

In particular, for $q = 0$ the equation is

$$(Y'Z' + Z'X' + X'Y')^2 - 4X'Y'Z'(X'+Y'+Z') = 0,$$

which is right, since, in the case in question (the tricuspidal curve), we have

$$X,\ Y,\ Z = X',\ Y',\ Z'.$$

I remark, in passing, that, taking the equation to be

$$(Y'Z' + Z'X' + X'Y')^2 = m\,X'Y'Z'(X' + Y' + Z'),$$

we may write herein

$$\begin{aligned} Z' &= \tfrac{1}{2} - x, \\ X' &= \tfrac{1}{4} + \tfrac{1}{2}x - \tfrac{\sqrt{3}}{2}y, \\ Y' &= \tfrac{1}{4} + \tfrac{1}{2}x + \tfrac{\sqrt{3}}{2}y\,; \end{aligned}$$

[p. 9] where

$$x = \frac{2\sqrt{m(m-3)}}{9}\cos\theta - \frac{2(m-3)}{9}\cos 2\theta,$$

$$y = \frac{2\sqrt{m(m-3)}}{9}\sin\theta + \frac{2(m-3)}{9}\sin 2\theta,$$

which are the formulae for the description of the trinodal quartic as a unicursal curve.

I consider now the second position; viz. the horizontal plane now passes through the centre of the tetrahedron, and is parallel to two opposite edges. The equations of the nodal lines are here $(y = 0,\ z = 0)$, $(z = 0,\ x = 0)$, $(x = 0,\ y = 0)$; and if for convenience we assume the distance of the mid-points of opposite edges to be $= 2$, or the half of this $= 1$, then the equations of the faces are

$$\begin{aligned} X &= x + y + z - 1 = 0, \\ Y &= -x - y + z - 1 = 0, \\ Z &= x - y - z - 1 = 0, \\ W &= -x + y - z - 1 = 0, \end{aligned}$$

and the equation of the surface is

$$\sqrt{X} + \sqrt{Y} + \sqrt{Z} + \sqrt{W} = 0.$$

Proceeding to rationalise, this is

$$X + Y + 2\sqrt{XY} = Z + W + 2\sqrt{ZW},$$

viz.

$$2z + \sqrt{XY} = \sqrt{ZW}\,;$$

we thence have

$$4z^2 + 4z\sqrt{XY} + XY = ZW,$$

or, since

$$ZW - XY = 4z + 4xy,$$

this is

$$z + xy - z^2 = z\sqrt{XY}\,;$$

whence

$$(z + xy - z^2)^2 = z^2\{(z-1)^2 - (x+y)^2\}\,;$$

or reducing,

$$x^2y^2 + z^2x^2 + z^2y^2 + 2xyz - z^2 = 0,$$

a form which puts in evidence the nodal lines. Considering z as constant, we have the equation of the section; this is a quartic having the node ($x = 0$, $y = 0$), and two other nodes at infinity on the two axes respectively; moreover, the curve has for bitangents the intersections of its plane with the faces of the tetrahedron; or what is the same thing, attributing to z its constant value, the equations of the bitangents are

$$\begin{aligned} x + y + z - 1 &= 0, \\ -x - y + z - 1 &= 0, \\ x - y - z - 1 &= 0, \\ -x + y - z - 1 &= 0. \end{aligned}$$

[p. 10] These lines form a rectangle which is the section of the tetrahedron; observe that this is inscribed in the square the corners of which are $x = \pm 1$, $y = \pm 1$; viz. $z = +1$ (highest section), this is the dexter diagonal (considered as an indefinitely thin rectangle), and as z diminishes, the longer side decreases and the shorter increases until for $z = 0$ (central section) the rectangle becomes a square; after which, for z negative it again becomes a rectangle in the conjugate direction, and finally, for $z = -1$ (lowest section) it becomes the sinister diagonal (considered as an indefinitely thin rectangle). But on account of the symmetry it is sufficient to consider the upper sections for which z is positive. The sides $\pm(x + y) + z - 1 = 0$ parallel to the dexter diagonal of the square may for convenience be termed the dexter sides, and the others the sinister sides. In what follows I write c to denote the constant value of z.

We require to know whether the bitangents have real or imaginary contacts; and for this purpose to find the coordinates of the points of contact.

Take first a dexter bitangent $x + y + c - 1 = 0$; the coordinates of any point hereof are

$$x = \tfrac{1}{2}(1 - c + \theta), \quad y = \tfrac{1}{2}(1 - c - \theta),$$

where θ is arbitrary; and substituting in the equation of the curve, we should have for θ a twofold quadric equation, giving the values of θ for the two points of contact respectively. We have

$$x^2 + y^2 = \tfrac{1}{2}\{(1 - c)^2 + \theta^2\}, \quad xy = \tfrac{1}{4}\{(1 - c)^2 - \theta^2\},$$

and thence

$$8c^2\{(1 - c)^2 + \theta^2\} + 8c\{(1 - c)^2 - \theta^2\} + \{(1 - c)^2 - \theta^2\}^2 = 0,$$

viz. this equation is

$$\{\theta^2 - (1 - c)(1 + 3c)\}^2 = 0,$$

a twofold quadric equation, as it should be; and the values of θ being $= \pm\sqrt{(1 - c)(1 + 3c)}$, we see that these, and therefore the contacts, are real from $c = 1$ to $c = -\frac{1}{3}$.

In exactly the same way for a sinister bitangent $\pm(x - y) - c - 1 = 0$, we have

$$x = \tfrac{1}{2}(1 + c + \phi), \quad -y = \tfrac{1}{2}(1 + c - \phi), \quad \text{and} \quad \phi = \pm\sqrt{(1 - 3c)(1 + c)},$$

viz. the values of ϕ, and therefore the contacts, are real from $c = \frac{1}{3}$ to $c = -1$.

That is,

$c = 1$ to	Contacts of Dexter Bitangents.	Contacts of Sinister Bitangents.
$c = 1$ to $\frac{1}{3}$	real,	imaginary,
$c = \frac{1}{3}$ to $-\frac{1}{3}$	real,	real,
$c = -\frac{1}{3}$ to -1	imaginary,	real.

or say $c = 1$ to $\frac{1}{3}$, the contacts are real, imaginary; but $c = \frac{1}{3}$ to 0, they are real, real. In the transition case, $c = \frac{1}{3}$, the sinister bitangents become osculating (4-pointic) tangents touching at points on the dexter diagonal. This can be at once verified.

[p. 11] Observe that when $c = 1$, we have

$$(x + y)^2 + x^2y^2 = 0,$$

so that the only real point is $x = 0$, $y = 0$; viz. this is a tacnode, having the real tangent $x + y = 0$. For $c = 0$ (central section) the equation becomes $x^2y^2 = 0$; viz. the curve is here the two nodal lines each twice.

It is now easy to trace the changes of form.

$c = 1$; curve is a tacnode, as just mentioned, tangent the dexter diagonal.

$c < 1 > \frac{1}{3}$; curve is a figure of 8 inside the rectangle, having real contacts with the dexter sides, but imaginary contacts with the sinister sides.

$c = \frac{1}{3}$; curve is a figure of 8 having real contacts with the dexter sides, and osculating (4-pointic) contacts with the sinister sides.

$c < \frac{1}{3} > 0$; curve is an indented figure of 8 having real contacts as well with the sinister as the dexter sides.

$c = 0$; curve is squeezed up into a finite cross, being the crunodal parts of the nodal lines; and joined on to these we have the acnodal parts, so that the whole curve consists of the lines $x = 0$, $y = 0$ each as a twofold line.

For tracing the curve, it is convenient to turn the axes through an angle of 45°; viz. writing $\dfrac{y+x}{\sqrt{2}}$, $\dfrac{y-x}{\sqrt{2}}$ in place of x, y respectively, the equation becomes

$$c^2(y^2 - x^2)^2 + c(y^2 + x^2) + \tfrac{1}{4}(y^2 - x^2)^2 = 0;$$
$$x = 0 \text{ gives } y^2 = 0 \text{ or } y^2 = -4c(1 + c),^*$$
$$y = 0 \text{ gives } x^2 = 0 \text{ or } x^2 = 4c(1 - c).$$

Moreover, we have

$$\tfrac{1}{4}(c - 1)^2(y^2 - x^2) + 8c^2y^2 + (y^2 - x^2)^2 = 0,$$

viz.

$$(x^2 + y^2 - 2c^2 - 2c)^2 = 4c^2\{(c - 1)^2 - 2y^2\},$$

and similarly

$$(x^2 + y^2 + 2c^2 - 2c)^2 = 4c^2\{(c + 1)^2 - 2x^2\},$$

putting in evidence the bitangents, now represented by the equations $c - 1 = \pm y\sqrt{2}$ and $c + 1 = \pm x\sqrt{2}$ respectively. And for the first of these, or $c - 1 = \pm y\sqrt{2}$, we have for the points of contact $x^2 = \frac{1}{2}(1 - c)(1 + 3c)$; and for the second of them, or $c + 1 = \pm x\sqrt{2}$, the points of contact are $y^2 = \frac{1}{2}(1 + c)(1 - 3c)$.

I consider the circumscribed cone having its vertex at a point $(0, 0, \gamma)$ on the nodal line ($x = 0,\ y = 0$). Writing in the equation of the surface $x = \lambda(z - \gamma)$, $y = \mu(z - \gamma)$, the equation, throwing out the factor $(z - \gamma)^2$, becomes

$$\lambda^2\mu^2(z - \gamma)^2 + 2\lambda\mu z(z - \gamma) + (\lambda^2 + \mu^2)z^2 - z^2 = 0,$$

that is,

$$(\lambda^2\mu^2 + \lambda^2 + \mu^2)\, z^2 + 2(-\gamma\lambda\mu + 1)\, z\lambda\mu + \gamma^2\lambda^2\mu^2 = 0\,;$$

* y always imaginary when c is positive.

[p. 12] and equating to zero the discriminant in regard to z, we have

$$\gamma^2\lambda^2\mu^2(\lambda^2\mu^2 + \lambda^2 + \mu^2) - (-\gamma\lambda\mu + 1)^2 = 0,$$

that is,

$$\gamma^2(\lambda^2 + \mu^2) + 2\gamma\lambda\mu - 1 = 0\,;$$

and substituting herein the values $\lambda = \dfrac{x}{z-\gamma}$ and $\mu = \dfrac{y}{z-\gamma}$, we have the equation of the cone, viz. this is

$$\gamma^2(x^2 + y^2) + 2\gamma xy - (z-\gamma)^2 = 0,$$

or, what is the same thing,

$$\gamma^2(x^2 + y^2 - 1) + 2\gamma(xy + z) - z^2 = 0\,;$$

viz. this is a quadric cone having for its principal planes $z = \gamma$, $x + y = 0$, $x - y = 0$, these last being the planes through the nodal line and the two edges of the tetrahedron.

In the particular case $\gamma = \infty$, the cone becomes the circular cylinder $x^2 + y^2 - 1 = 0$.

The cone intersects the plane $z = 0$ in the conic

$$\gamma^2(x^2 + y^2 - 1) + 2\gamma xy = 0,$$

which is a conic passing through the corners of the square $(x = 0,\ y = \pm 1)$, $(x = \pm 1,\ y = 0)$. For $\gamma > 1$, that is, for an exterior point, the conic is an ellipse having for the squares of the reciprocals of the semi-axes $1 + \dfrac{1}{\gamma}$, $1 - \dfrac{1}{\gamma}$ (this at once appears by writing in the equation $\dfrac{x+y}{\sqrt{2}}$, $\dfrac{x-y}{\sqrt{2}}$ in place of x, y respectively). In particular, for $\gamma = \infty$, the curve becomes the circle $x^2 + y^2 = 1$. We have thus the apparent contour of the surface as seen from the point $z = \gamma$ on the nodal line, projected on the plane $z = 0$ of the other two nodal lines.

To find the curve of contact of the cone and surface, or say the surface-contour from the same point, write for a moment

$$V = \gamma^2(x^2 + y^2 - 1) + 2\gamma(xy + z) - z^2,$$

$$U = (xy+z)^2 - z^2(x^2+y^2-1),$$

then, substituting for x^2+y^2-1 its value in terms of V from the first equation, we find

$$U = \left(xy+z-\frac{z^2}{\gamma}\right)^2 \frac{V}{\gamma^2},$$

and the equations $V=0$, $U=0$ give therefore

$$xy+z-\frac{z^2}{\gamma}=0,$$

or say $\gamma(xy+z)-z^2=0$.

The cone and surface therefore touch along the quadriquadric curve

$$\gamma^2(x^2+y^2-1)+2\gamma(xy+z)-z^2=0,$$

$$\gamma(xy+z)-z^2=0,$$

equations which may be replaced by

$$\gamma^2(x^2+y^2-1)+xy+z=0,$$

$$\gamma^2(x^2+y^2-1)+z^2=0.$$

In the case $\gamma=\infty$, the equations are $x^2+y^2-1=0$, $xy+z=0$, viz. the curve is the intersection of the hyperbolic paraboloid $xy+z=0$ by the cylinder $x^2+y^2-1=0$.

557. On Certain Constructions for Bicircular Quartics

[p. 13] *[From the Proceedings of the London Mathematical Society, vol. v. (1873–1874), pp. 29–31. Read March 12, 1874.]*

I CALL to mind that if F, G are any two points and F', G' their antipoints, then the circle on the diameter FG and that on the diameter $F'G'$ are concentric orthotomics, viz. they have the same centre, and the sum of the squared radii is $=0$. Moreover, if the circles B, B' are concentric orthotomics, and the circle A is orthotomic to B, then it is a bisector of B', viz. it cuts B' at the extremities of a diameter of B'; and B is then said to be a bifid circle in regard to A.

Given two real circles, these have an axial orthotomic, the circle, centre on the line of centres at its intersection with the radical axis, which cuts at right angles the given circles; viz. this axial orthotomic is real if the circles have no real intersection; but if the intersections are real, then the axial orthotomic is a pure imaginary, and instead thereof we may consider its concentric orthotomic, viz. this is the axial bifid of the two circles, or circle having its centre on the line of centres at the intersection thereof with the radical axis or common chord of the two circles, and having this common chord for its diameter.

If one of the circles is a pure imaginary, then we have still an axial orthotomic; viz. the pure imaginary circle is replaced by the concentric orthotomic; and the axial orthotomic is a bisector of the substituted circle; and so if each of the circles is a pure imaginary, then we have still an axial orthotomic, viz. each circle is replaced by the concentric orthotomic, and the axial orthotomic is a bisector of the substituted circles. And in either case the axial orthotomic of the original circles (one or each of them pure imaginary) is real; viz. this is given either as the axial bisector of one real circle and orthotomic of another real circle; or as the axial bisector of two circles, from which the reality thereof easily appears. Or we may verify it thus: Suppose

[p. 14] that the two circles are $(x-\alpha)^2+y^2=\beta^2$, $(x-\alpha')^2+y^2=\beta'^2$, and their axial orthotomic $(x-m)^2+y^2=k^2$, then we have $(m-\alpha)^2=\beta^2+k^2$, $(m-\alpha')^2=\beta'^2+k^2$; subtracting, it appears that m is real; and then if either β^2 or β'^2 is negative, the equation containing this quantity shows that k^2 is positive; viz. the circle $(x-m)^2+y^2=k^2$ is real.

The above remarks have an obvious application to the theory of bicircular quartics; viz. a bicircular quartic is the envelope of a variable circle, having its centre on a conic, and orthotomic to a circle: it may be that this circle is a pure imaginary. We then replace it by the concentric orthotomic, and say that the curve is the envelope of a variable circle having its centre on a conic and bisecting a circle. We have thus a real form for cases which originally present themselves under an imaginary form.

The Bicircular Quartic with given vertices.

First, if the vertices are real; let the vertices taken in order be F, G, H, K.

First construction: On HK as diameter describe a circle, and on FG as diameter a circle; on the line terminated by the two centres (as transverse or conjugate axis) describe a conic Θ_1, and describe the axial orthotomic circle Σ_1 of the two circles (viz. the centre of Σ_1 is on the axis of symmetry at its intersection with the radical axis of the two circles); then the curve is the envelope of a variable circle having its centre on Θ_1 and orthotomic to Σ_1.

Second construction: On FH as diameter describe a circle, and on GK as diameter a circle. On the line terminated by the two centres (as transverse or conjugate axis) describe a conic Θ_2, and describe the axial bifid circle Σ_2' of the two circles (viz. the centre of Σ_2' is on the axis of symmetry at its intersection with the radical axis or common chord of the two circles, and its diameter is this common chord); then the curve is the envelope of a variable circle having its centre on Θ_2 and bisecting Σ_2'.

Third construction: On FK as diameter describe a circle, and on GH as diameter a circle; and then, as in the first construction, a conic Θ_3 and a circle Σ_3; the curve is the envelope of a variable circle having its centre on Θ_3 and orthotomic to Σ_3.

Observe that in the three constructions the conics have always the same centre; and if the three conics are taken with the same foci, then the three constructions give one and the same bicircular quartic. The first and third constructions form a pair, and there is no reason for selecting one of them in preference to the other; but the second construction is unique; it is on this account natural to make use of it in discussing the series of curves with the given vertices.

In the particular case where the points F, G and H, K are situate symmetrically on opposite sides of a centre O ($OF = OK$, $OG = OH$), then in the third construction the centres each coincide with O, or the axis of the conic vanishes; hence the construction fails: the first and second constructions hold good, and in each of them the

[p. 15] conic and circle are concentric. The curve is in this case quadrantal: having, besides the original axis of symmetry, another axis of symmetry through O, at right angles thereto.

Secondly, if the vertices are two real, two imaginary, say f, $g = \alpha \pm \beta i$; h, k, we modify the first or third construction; viz. if F', G' are the antipoints of F, G; then on $F'G'$ as diameter describe a circle, and on HK as diameter a circle. On the line terminated by the two centres (as transverse or conjugate axis) describe a conic Θ_1, and describe the axial bisector-orthotomic circle Σ_1 of the two circles; viz. this is the circle (centre on the axis of symmetry) which bisects the circle $F'G'$, and cuts at right angles the circle HK; then the curve is the envelope of the variable circle having its centre on Θ_1 and orthotomic to Σ_1.

Thirdly, if the vertices are all imaginary, say f, $g = \alpha \pm \beta i$; h, $k = \gamma \pm \delta i$, we modify the first or third construction. Take F', G' the antipoints of F, G, and H', K' the antipoints of H, K; then on $F'G'$ as diameter describe a circle, and on $H'K'$ as diameter a circle; on the line terminated by the two centres (as transverse or conjugate axis) describe a conic Θ, and describe the axial bisector-circle Σ of the two circles (viz. this is a circle, centre on the axis of symmetry, bisecting each of the circles); the curve is the envelope of a variable circle, centre on the conic Θ and cutting at right angles the circle Σ.

558. A Geometrical Interpretation of the Equations Obtained by Equating to Zero the Resultant and the Discriminants of Two Binary Quantics

[p. 16] *[From the Proceedings of the London Mathematical Society, vol. v. (1873–1874), pp. 31–33. Read March 12, 1874.]*

CONSIDER the equations

$$U = (a,\ b,\ \dots \mid t,\ 1)^{\lambda} = 0,$$

$$U' = (a',\ b',\ \dots \mid t,\ 1)^{\lambda'} = 0\,;$$

and equating to zero the discriminants of the two functions respectively, and also the resultant of the two functions, let the equations thus obtained be

$$\Delta = (a,\ b,\ \dots)^{2\lambda-2} = 0,$$

$$\Delta' = (a',\ b',\ \dots)^{2\lambda'-2} = 0,$$

$$R = (a,\ b,\ \dots)^{\lambda'}(a',\ b',\ \dots)^{\lambda} = 0.$$

I take $(a,\ b,\ \dots)$, $(a',\ b',\ \dots)$ to be linear functions of the coordinates $(x,\ y,\ z)$; and t to be an indeterminate parameter. Hence $U = 0$ represents a line the envelope whereof is the curve $\Delta = 0$, or, what is the same thing, the equation $U = 0$ represents any tangent of the curve $\Delta = 0$; this is a unicursal curve of the order $2\lambda - 2$ and class λ, with $3(\lambda - 2)$ cusps and $\frac{1}{2}(\lambda - 2)(\lambda - 3)$ nodes. Similarly $U' = 0$ represents a line the envelope of which is the curve $\Delta' = 0$: this is a unicursal curve of the order $2\lambda' - 2$ and class λ', with $3(\lambda' - 2)$ cusps and $\frac{1}{2}(\lambda' - 2)(\lambda' - 3)$ nodes; the equation $U' = 0$ represents any tangent of this curve.

The equations $U = 0$, $U' = 0$ considered as existing simultaneously with the same value of t, establish a $(1,\ 1)$ correspondence between the tangents (or if we please, between the points) of the two curves. The locus of the intersection of the corresponding tangents is the curve $R = 0$, a unicursal curve of the order $\lambda + \lambda'$, with $\frac{1}{2}(\lambda + \lambda' - 1)(\lambda + \lambda' - 2)$ nodes and no cusps; consequently of the class $2(\lambda + \lambda' - 1)$.

[p. 17] It is to be shown that the curve $R = 0$ touches the curve $\Delta = 0$ in $\lambda' + 2\lambda - 2$ points, and similarly the curve $\Delta' = 0$ in $2\lambda' + \lambda - 2$ points.

In fact, consider any tangent T' of the curve Δ'; let this meet the curve Δ in a point A, and let Q be the tangent at A to the curve Δ; suppose, moreover, that T is the tangent of Δ corresponding to the tangent T' of Δ'. Then if Q and T coincide, the corresponding tangent of T' will be Q, and the curve R will pass through A. It is easy to see that in this case the curves R, Δ will touch at A.

Again, if P be a tangent from A to the curve Δ, then, if T and P coincide, the corresponding tangent of T' will be P, and the curve R will pass through A; but in this case the point A will be a mere intersection, not a point of contact, of the two curves.

The tangents T, Q each correspond to T', and they consequently correspond to each other. For a given position of T we have a single position of T', and therefore $2\lambda-2$ positions of A, or, what is the same thing, of Q; that is, for a given position of T we have $2\lambda-2$ positions of Q. Again, to a given position of Q corresponds a single position of A, therefore λ' positions of T', therefore also λ' positions of T; that is, for a given position of Q we have λ' positions of T. The correspondence between T, Q is thus a $(\lambda',\ 2\lambda-2)$ correspondence, and the number of united tangents is therefore $\lambda'+2\lambda-2$, or the curves R, Δ touch in $\lambda'+2\lambda-2$ points.

The tangents T, P each correspond to T', and they therefore correspond to each other. For a given position of T we have a single position of T', and therefore $2\lambda-2$ positions of A, and thence $(2\lambda-2)(\lambda-2)$ positions of P; that is, for a given position of T we have $(2\lambda-2)(\lambda-2)$ positions of P. Again, to a given position of P correspond $2\lambda-4$ positions of A, therefore $(2\lambda-4)\lambda'$ positions of T' or of T; that is, for a given position of P we have $(2\lambda-4)\lambda'$ positions of T. The correspondence between T, P is thus a $[2\lambda'(\lambda-2),\ 2(\lambda-1)(\lambda-2)]$ correspondence, and the number of united tangents is $2(\lambda+\lambda'-1)(\lambda-2)$; or the curves R, Δ meet in $2(\lambda+\lambda'-1)(\lambda-2)$ points.

Reckoning the contacts twice, the total number of intersections of R, Δ is

$$2\lambda'+4\lambda-4+2(\lambda+\lambda'-1)(\lambda-2)=(\lambda+\lambda')(2\lambda-2),$$

as it should be.

In the particular case $\lambda=\lambda'=2$, the curves Δ, Δ' are conics, and the curve R is a quartic curve touching each of the conics 4 times; this is at once verified, since the equations here are

$$ac-b^2=0,\quad a'c'-b'^2=0,\quad 4(ac-b^2)(a'c'-b'^2)-(ac'+a'c-2bb')^2=0.$$